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ORIGINAL ARTICLE



Occurrence of potentially traumatic events, type, and severity in undergraduate students

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ABSTRACT

Background: Trauma exposure can have significant mental health impacts on students that impair academic performance and engagement. It is therefore important to describe the occurrence of trauma experiences and associated symptoms among undergraduate students.

Objective: This study aimed to 1) assess the rates of potentially traumatic experiences, 2) examine the frequency of different trauma experience types, and 3) compare post-traumatic stress (PTS) symptom severity across experience subtypes in a large sample of undergraduate students.

Methods: Participants were 806 undergraduate students from three Australian universities who completed online self-report measures of trauma exposure, psychological distress, and PTS symptoms.

Results: Approximately two-thirds of students (64%) reported having experienced at least one potentially traumatic event. The most common experience types were the unexpected death of a loved one and accidents. Students reporting exposure to potential trauma had significantly higher general distress than non-exposed students. Events involving interpersonal violence were associated with greater PTS symptom severity.

Conclusions: This study provides evidence that the occurrence of potentially traumatic events is high among university students and may carry negative effects for mental health and functioning. Results highlight the need for trauma-informed educational approaches and robust counselling services to support students managing trauma histories.

KEY POINTS

What is already known about this topic:

- (1) Trauma occurrence in undergraduates affects mental health and academic outcomes.
- (2) Interpersonal violence trauma relates to higher psychological distress.
- (3) Existing research has focused on specific trauma subtypes or PTSD criteria, overlooking the broader spectrum of trauma experiences and their psychological impacts.

What this topic adds:

- (1) This study provides updated evidence of potential trauma experiences among university students in Australia.
- (2) Interpersonal violence was consistently linked to greater post-traumatic stress symptom severity, supporting the need for specialised support services.
- (3) Findings emphasise the necessity for trauma-informed educational approaches and robust counselling services to aid students with trauma histories.

ARTICLE HISTORY

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KEYWORDS

Trauma; trauma prevalence; posttraumatic stress disorder; psychological distress

University studies can be a stressful experience for many students. This is evident through substantially higher levels of psychological distress experienced by undergraduate university students when compared to other samples of similar age (Baik et al., 2019; Sharp & Theiler, 2018). The transition into adulthood is generally associated with numerous stressors, including

changes to living arrangements, relationships, and employment, which, for many students, co-occur with the transition to university (Macaskill, 2013; Matud et al., 2020). For some, such changes can generate instability and uncertainty, contributing to a developmentally and psychologically critical period (Matud et al., 2020; Pedrelli et al., 2015). Furthermore,

this period also corresponds with the peak age for developing mental health disorders (Auerbach et al., 2018; Huang et al., 2018; Pedrelli et al., 2015). Given this confluence of stressors, it is little surprise that high levels of psychological distress are prevalent during this phase of life (Auerbach et al., 2018; Pedrelli et al., 2015; Sharp & Theiler, 2018).

Students who have experienced a traumatic event may be particularly prone to high levels of psychological distress (Frazier et al., 2009; Smyth et al., 2008). Traumatic events can be conceptualised as an event or experience involving actual or threatened death, serious injury, or violence/abuse (Chafouleas et al., 2019; Perfect et al., 2016). The experience of traumatic events has been associated with significant impairments in general functioning and can trigger extreme distress (Forbes et al., 2019; Muldoon et al., 2019; Perfect et al., 2016). Such emotional states have been associated with problematic health behaviours, including lack of concentration, indecisiveness, changes in appetite and sleeping, and reduction in motivation, memory, and recall (Huang et al., 2018; Kirk & Dollar, 2002).

Previous studies have investigated trauma prevalence and PTS symptoms; however, they have tended to examine only certain subtypes of traumas (e.g., childhood trauma or sexual abuse) or have limited the events to only those that meet the PTSD diagnosis criteria (Carey et al., 2018; Duncan, 2000; Kelley et al., 2009). While understanding the impacts of PTSD is important, the vast majority of people exposed to potentially traumatic events experience some persisting symptoms that, although not meeting the diagnostic criteria for PTSD (Boyraz et al., 2016; Frazier et al., 2009; Smyth et al., 2008), limit their ability to function optimally, impacting their social, emotional, and psychological well-being (Frieze, 2015). This highlights the importance of examining the wider psychological effects of exposure to potentially traumatic events beyond those meeting specific PTSD diagnosis criteria.

It has been well established that individuals with a history of trauma often exhibit poorer social, emotional, and cognitive functioning (Malarbi et al., 2017; Perfect et al., 2016; Van Der Kolk, 2007). Given the many noted adverse effects, trauma experience is likely to pose additional major challenges to performing well at university. A substantial body of literature suggests that university students with posttraumatic stress disorder tend to show lower academic performance, poorer physical health, and higher psychological distress (Cusack et al., 2019; Im et al., 2020; Lawler et al., 2005). It is therefore important to understand not only the frequency and types of potentially traumatic experiences observed across in this population but

also to determine how different events manifest in both clinical and sub-clinical levels of PTS symptoms.

Previous studies investigating exposure to trauma and symptoms in university samples have compared PTSD symptom profiles across different trauma types, often limited to a few common experiences, to determine which events are associated with greater distress. For example, Kelley et al. (2009) compared three commonly reported trauma types, finding that those who had experienced sexual assault exhibited significantly higher symptom severity than those involved in a motor vehicle accident, or those exposed to sudden or unexpected death. This suggests that certain event types, such as sexual assault, are associated not only with greater risk for a PTSD diagnosis but also with higher levels of overall PTSD symptom severity (Kelley et al., 2009). Duncan (2000) followed 210 university students over time, comparing enrolment rates among those with and without a history of childhood abuse. They observed higher dropout among those with a history of childhood abuse compared to those with no trauma history (Duncan, 2000). While these studies demonstrate some of the symptom profiles and effects of differences in trauma types, few studies have examined the broader prevalence and impact of different trauma types in terms of PTS symptom severity.

One of the few studies to examine the occurrence of different types of potentially traumatic events and associated indicators of psychological distress was Frazier et al. (2009). They examined these in a large undergraduate sample, finding that the vast majority of students (85%) had experienced a potentially traumatic event in their lifetime. The study revealed that the unexpected death of a loved one was the most frequently reported event and that experiences of family violence, unwanted sexual attention, and sexual assault were linked to higher levels of current distress. While the study provides important insights regarding the occurrence of potentially traumatic events and event types in an undergraduate sample, findings are obviously limited to both the specific US population examined, and the timeframe. It is important therefore to update such findings and provide insights from different populations.

The current study seeks to extend findings on trauma exposure and associated patterns of distress in university students by examining both the severity of PTS symptoms and the frequency of exposure to potentially traumatic events within a large Australian undergraduate sample. In line with this, the aim of the present study is to describe the occurrence of exposure to potentially traumatic events (risk factors

for distress) and associated patterns of PTS symptoms by event type, and psychological distress (actual experience of distress) in this undergraduate group. Specifically, the study sought to: i. describe the occurrence of exposure to potentially traumatic experiences and associated psychological distress symptoms among undergraduate students, comparing those with exposure to those without, ii. detail the common types of experiences, and iii. examine the severity of PTS symptoms across different experience subtypes, focusing on the single most severe event endorsed by each individual. For the present study, we focused on identifying the proportion of individuals who have experienced potentially traumatic events and the type of event, focusing on the single most severe event experience reported by any individual. This study was completed as part of a multi-site research project investigating the relationships between patterns of cognition and emotion in undergraduate students. Self-report measures of exposure to potentially traumatic events, psychological distress, and PTS symptoms were administered to a large undergraduate sample to examine the association between event types and the experienced PTS symptoms.

Materials and methods

Participants

The sample included undergraduate psychology students from three Australian Universities, Curtin University, The University of Western Australia, and the University of Sydney. These psychology units are among the largest unit enrolments within the respective institutions due to the high number of students from multiple disciplines who enroll in these as electives. While these units have a high concentration of individuals enrolled in psychology, they include a substantial cross-section of other disciplines. Participants accessed the study via the university's undergraduate recruitment pool and participated for course credit. The research platform allows prospective undergraduate participants to view and select studies to be involved in where students can view study information and sign up to participate for course credit. No exclusion criteria were applied. The final sample comprised 806 participants (female = 581, male = 219, nonbinary = 6), ranging in age from 16 to 58 years ($M = 20.7$, $SD = 4.91$), with 224, 313, 340 from Curtin University, the University of Sydney, and the University of Western Australia respectively. More detailed demographic descriptions are provided in the results section.

Measures

Psychological distress: Depression, Anxiety, and Stress Scale (21-item)

The 21-item Depression Anxiety Stress Scale (DASS; Lovibond & Lovibond, 1995) was employed to measure the symptoms of psychological distress experienced by participants, providing a comprehensive assessment of their psychological state following trauma exposure. This comprises three subscales assessing the severity of symptoms associated with depression, anxiety, and stress. Participants rate the degree to which each statement applied to them over the past week on a four-point Likert scale from 0 (*Did not apply to me*) to 3 (*Applied to me very much or most of the time*), with higher scores representing greater symptom severity. Previous studies have shown the DASS-21 to be a valid and reliable measure among university students, indicating moderate-to-high internal consistency ($\alpha = .91-.93$, $\alpha = .81-.86$, and $\alpha = .86-.88$ for depression, anxiety, and stress, respectively; Baik et al., 2019; Coker et al., 2018). The internal consistency in the current sample was observed to be high across subscale and total score measures: $\alpha = .91$ (Depression), $\alpha = .86$ (Anxiety), $\alpha = .87$ (Stress), $\alpha = .94$ (Total).

Trauma screening and categorisation: Traumatic Events Questionnaire

The Traumatic Events Questionnaire (TEQ; Lauterbach & Vrana, 2001) was used to screen for exposure to potentially traumatic events, which are considered risk factors for developing psychological distress. The TEQ is an 11-item self-report measure assessing whether participants have been exposed to different categories of potentially traumatic events. These include accidents, natural disasters, violent crime, child abuse, adult abusive experiences, witnessing the death/injury of someone, being in a dangerous or life-threatening situation, and receiving news of the unexpected or sudden death of a loved one. Using a two-point dichotomous scale (*yes or no*), the first nine items ask participants to indicate whether they have or have not experienced the particular event. For example, "As a child, were you the victim of either physical or sexual abuse?". The final two items inquire about experiences with potentially traumatic events not previously mentioned, or if there are events that the individual prefers not to disclose. Participants who answered "yes" to any of the 11 questions they are asked to provide a brief description (a few words) describing the most troublesome event (e.g., "car accident"). In a sample of students, the TEQ has been

shown to have high reliability ($\alpha = .91$) for the number of events experienced in a two-week test-retest interval (Lauterbach & Vrana, 2001).

Participants who identified more than one event were asked to indicate their most severe experience endorsed and provided a brief description of this. Trauma symptoms (PCL) was then completed in relation to this event. The allocation of different experiences to categories was guided by responses to the TEQ questions as detailed in Table 1. This resulted in 9 designated categories and a 10th category for potentially traumatic events not specified. Responses to questions 4 and 6 were divided into sub-types of adult or child abusive experiences (Sexual Abuse/ Assault, Emotional Abuse, Physical Abuse, Domestic Violence) based on the description provided of the nature of the most severe experience. This therefore resulted in nine over-arching categories with four sub-categories each for adult and child abusive experiences.

Trauma symptoms measure: Post-Traumatic Stress Disorder Checklist – Civilian

The Post-Traumatic Stress Disorder Checklist, Civilian version (PCL-C; Weathers et al., 1993) was used as the key measure of distress experienced by participants relating specifically to traumatic stress symptom severity stemming from their single most severe experience. The PCL-C contains 17 items reflecting DSM-IV symptoms of PTSD within three subscales corresponding to re-experiencing, avoidance and hyperarousal symptoms. Participants indicate the degree to which each symptom has bothered them within the past month on a five-point Likert scale from 1 (*not at all*) to 5 (*extremely*). Examples of items include statements such as “Repeated, disturbing memories, thoughts, or

images of a stressful experience from the past?”. Items are summed to measure total symptom severity (range = 17 to 85). A score of 30–43 indicates a moderate to moderately high presence of PTSD symptoms, while a score of 44 or higher is suggestive of a probable PTSD diagnosis according to DSM-IV criteria (Norris & Hamblen, 2004). The PCL-C has been shown to have high internal consistency ($\alpha = .94$), and correlates strongly (i.e., $r > .75$) with other PTSD symptomatology measures (Ruggiero et al., 2003). Internal consistency for the current sample was high ($\alpha = .94$).

Procedure

The project received ethical approval from each institution’s Human Research Ethics Committee (approval# 2021/ET000074). Participants accessed the study via the University’s undergraduate recruitment pool. This is a research platform that allows prospective undergraduate participants to view and select studies to be involved in where students can view study information and sign up to participate for course credit. The study did not specifically advertise for recruitment of individuals with trauma experience. However, the information form did indicate that the study would involve questions relating to the experience of potentially traumatic events. Upon registering interest in the study, participants followed a link to Inquisit 6 (2022) which managed the delivery of the study. Participants provided initial informed consent before commencing the study. They then completed a brief demographics questionnaire, including gender, age, and ethnicity (to permit potential comparison between samples). This was followed by a battery of cognitive and emotional measures, including the DASS-21, TEQ, and PCL-C. A branching function was included such that only

Table 1. TEQ question items and corresponding event categories adopted.

	TEQ Question	Category
1	Serious industrial, farm or car accidents or fire/explosion	Transportation Accident
2	Natural disasters	Natural disasters
3	Violent crime/Robbery	Crime–Robbery
4	As a Child–Victim of physical or sexual abuse	Childhood abusive experience 1. Sexual Abuse/Assault 2. Emotional Abuse 3. Physical Abuse 4. Domestic Violence
5	As an Adult – unwanted sexual experience	Adult abusive experience 1. Sexual Abuse/Assault
6	Adult abuse, physical or otherwise	Adult abusive experience 2. Emotional Abuse 3. Physical Abuse 4. Domestic Violence
7	Witnessed someone mutilated, seriously injured, or violently killed	Witnessed Trauma
8	Been in serious danger of losing life or serious injury	Severe Illness/Injury
9	Received news of mutilation, serious injury, or violent/unexpected death of someone close	Received news
10	Any other traumatic event	Unspecified Event

those who indicated having experienced one or more potentially traumatic events on the TEQ completed the PCL-C. This study is part of a broader Cognition Emotion Research Collaboration Initiative (CERCI) project which includes several emotional and cognitive components gathered across sites. Completion of the entire study took approximately one hour, after which participants received debrief information, which included contact details of university-based and other counselling services that they could access if they wished, in addition to the contact details of the researchers.

Results

Data screening

Data were initially screened and examined for missing values. Data management and analysis were performed using Jamovi (The jamovi project, 2023). The initial sample comprised 906 participants. Thirty-eight participants (4.19%) reported no demographics and were excluded. As all questionnaires were delivered in English, any participants who reported low English proficiency were excluded (24 participants, 2.65%). A further 25 cases (2.76%) were deleted due to large missing data (i.e., reporting demographics only). Any further participants who recorded missing data for critical measures were then excluded, with five participants (0.55%) missing all DASS-21 data and six (0.66%) missing PCL-C questionnaire. Two final participants (0.22%) were excluded due to responding that suggested a lack of engagement with the questionnaire measures (i.e., reporting the same response e.g., 1,1,1,1 ... for all items across all three measures). Following all exclusions ($N = 100$, 11.04%), the final sample included for analyses was 806 participants. Descriptive statistics by institution are provided in

Table 2. The current data (individual demographics removed) are available on the Open Science Framework via the following link: https://osf.io/9ja82/?view_only=11751bb9be4d4155b15ea2ae167f21a1

Frequency of potentially traumatic events experienced

Approximately two-thirds of the sample ($n = 516$, 64%) reported experiencing at least one potentially traumatic event in their lifetime, with the remainder reporting no trauma experience ($n = 290$, 36%). **Table 3** summarises the number of participants reporting each event type. Of the ten over-arching categories, the most frequent event experienced was "Received news" of a traumatic event followed by "Adult abusive experience". The remaining trauma categories from most frequently experienced to least frequently experienced were "Unspecified event", "Childhood abusive experience", "Transportation accident", "Severe illness/injury", "Threat of serious injury/danger", "Natural Disaster", "Witnessed Trauma", and "Crime – Robbery". The "Unspecified traumatic event" category included those who did not wish to report their most severe experience, or ambiguous descriptors of the event.

Severity of PTS symptoms across event types

The level of trauma symptom severity was examined as a function of the event types endorsed under each of the specific sub-categories detailed in **Table 3**. A One-Way ANOVA was conducted to compare the severity of trauma symptoms between these subtypes with the dependent variable being PCL-C scores and the independent variable being type of potentially traumatic experience. Results of the ANOVA revealed

Table 2. Number of participants, gender, mean age (SD), and ethnicity by institution and across the total sample.

	Institution			Total	
	Curtin	USyd	UWA		
Age – Years	22.5 (5.80)	20.5 (4.10)	19.9 (4.74)	20.7 (4.91)	
Gender F/M/NB	147/46/0	219/67/1	215/106/5	581/219/6	
				<i>n</i>	%
Total Participants	193	287	326	806	100
Ethnicity					
White	154	222	271	647	80.3
European	8	21	10	39	4.8
Middle Eastern	5	19	12	36	4.5
African	3	1	9	13	1.6
Mixed	19	18	19	56	6.9
Other	4	6	5	15	1.9

Note. F = Female, M = Male, NB = Non-binary; Curtin = Curtin University; USyd = University of Sydney; UWA = The University of Western Australia.

Table 3. Number of participants and sample proportion by event type with symptom severity in line with probable PTSD diagnosis.

Event Type/Subtype	N	%	Probable PTSD	
			N	%
Childhood Abusive Experience	54	10.5	27	50.0%
Sexual Abuse/Assault	31	6.0	16	51.6%
Emotional Abuse	9	1.7	4	44.4%
Physical Abuse	6	1.2	2	33.3%
Domestic Violence	8	1.6	5	62.5%
Adult Abusive Experience	115	21.4	53	46.1%
Sexual Abuse/Assault	36	7.0	18	50.0%
Emotional Abuse	40	7.8	20	50.0%
Physical Abuse	22	4.3	8	36.4%
Domestic Violence	17	3.3	7	42.1%
Severe Illness/Injury	35	6.8	21	60.0%
Received News	125	24.2	41	32.8%
Transportation Accident	42	8.1	7	16.7%
Threat of serious injury/danger	28	5.4	1	3.6%
Crime - Robbery	11	2.1	3	27.3%
Natural Disaster	18	3.5	4	28.6%
Witnessed Trauma	14	2.7	2	11.1%
Unspecified Event	74	14.3	44	59.5%

Note. Event subtype percentage represents the proportion of all individuals exposed to a potential traumatic event. Probable PTSD percentage represents the proportion of the subsample with symptom severity consistent with clinical cutoff.

a significant difference between the severity of trauma symptoms experienced across the different event types (Welch's $F(15, 90.3) = 7.84, p < .001, \eta^2 = .036$).

Figure 1 shows average PTSD mean values across each of the event subtypes.

Follow-up post hoc comparisons on the PCL-C scores were conducted using Games–Howell Post-Hoc test for multiple comparisons. The categories involving sexual/physical/emotional abuse/neglect (childhood abuse types, adult abuse/assault and domestic violence, severe illness/injury) did not differ from one another, showing similarly high levels of trauma symptoms. However, those that reported “Transportation Accident” as their most severe event experienced significantly lower levels of symptom severity than both the “Sexual abuse” ($M_{diff} = 14.46, p < .01$), and “Child sexual abuse” subtypes ($M_{diff} = 16.47, p < .01$), as well as the “Emotional abuse” ($M_{diff} = 13.62, p < .01$), “Child domestic violence” ($M_{diff} = 20.70, p < .05$), and the “Severe Illness/injury” subtypes ($M_{diff} = 14.74, p < .01$).

Those reporting “Threat of serious injury/danger” as their most severe event exposure experienced significantly lower levels of symptom severity than those indicating their most severe event as being “Sexual abuse” ($M_{diff} = 18.46, p < .001$), “Child sexual abuse” ($M_{diff} = 20.47, p < .01$), “Domestic violence” ($M_{diff} = 14.75, p < .05$), “Child domestic violence” ($M_{diff} = 20.70, p < .001$), as well as “Emotional abuse” ($M_{diff} = 17.62, p < .001$), “Physical abuse” ($M_{diff} = 13.52, p < .05$), and “Severe Illness/injury” ($M_{diff} = 18.74, p < .01$).

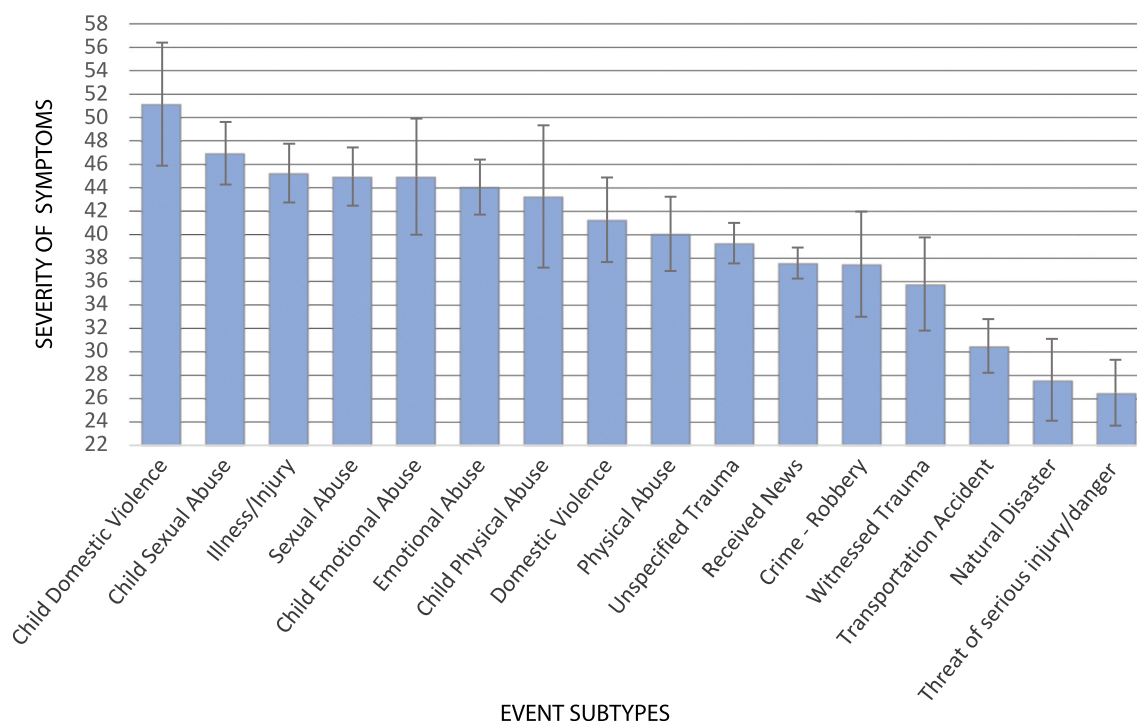


Figure 1. Mean PTSD symptoms (PCL-C) values across each event subtype, error bars depict standard error.

Those reporting “Natural Disasters” as their most severe event exposure experienced significantly lower levels of symptom severity than the “Sexual abuse” ($M_{diff} = 17.39$, $p < .05$), “Child sexual abuse” ($M_{diff} = 19.40$, $p < .05$), “Child domestic violence” ($M_{diff} = 23.62$, $p < .05$), and “Severe Illness/injury” subtypes ($M_{diff} = 17.67$, $p < .05$). Similarly, those reporting “Received news” as their most severe event experienced significantly lower levels of symptom severity than those in the “Threat of serious injury/danger” ($M_{diff} = 11.15$, $p < .001$), as did those who reported an “Unspecified event” ($M_{diff} = 12.79$, $p < .001$).

The pattern of these differences generally characterises lower-level traumatic symptoms observed for accidents, natural disasters, and those who felt in threat of a serious injury or dangerous situation, as compared to events where another individual was responsible for inflicting the experience (childhood and adult sexual/emotional/physical abuse, and domestic violence trauma types).

A follow-up examination of data using descriptors for the different levels of PCL-C severity levels indicated by Norris and Hamblen (2004) across the different event categories is also consistent with this interpretation. Specifically, as shown in Table 2, the highest proportion of those with PCL-C scores falling in the “probable PTSD” range (a score of 44 or higher) were among those who indicated any of the sub-types involving an event inflicted by another individual (child or adult Sexual/Emotional/Physical Abuse and Domestic Violence) as their worst experience, averaging 46.18% of the sample exposed to potentially traumatic events, while for the remaining event categories, the average proportion falling in the diagnostic range was 25.99%.

The relationship between exposure to potential trauma experience and psychological distress

An independent samples *t*-test was conducted to compare levels of psychological distress between those exposed to a potentially traumatic event and the non-exposed groups, with the dependent variable being general psychological distress as indicated by total scores on the DASS-21. Results revealed that the severity of psychological distress was indeed significantly higher (5.89 points, 95% CI [7.85, 3.92,]) for the event-exposed group ($M = 25.0$, $SD = 14.7$) compared to the non-exposed group ($M = 19.1$, $SD = 13.0$); $t(663) = 5.88$, $p < .001$, $d = 0.42$, 95% CI [.56, .27].

Discussion

This study sought to examine the nature of the relationship between the experience of potentially traumatic events and distress symptoms via three key aims: i. to describe the occurrence of potentially traumatic experiences and effects on psychological distress symptoms, ii. to detail the frequency and common event types experienced by undergraduates, and iii. to compare the severity of PTS symptoms across different event subtypes.

In assessing the first key aim, around two-thirds of the participants in the study (64%) reported having experienced at least one potentially traumatic event during their lifetimes. In examining the difference between the event-exposed and the non-trauma-exposed participants, those exposed to potentially traumatic events exhibited significantly higher levels of overall psychological distress compared to the non-trauma-exposed participants with a medium effect. This finding is broadly consistent with earlier research examining differences amongst students’ well-being in those with and without trauma exposure. However, the rates of event exposure in the current study appeared substantially lower (64% vs. 85%; Frazier et al., 2009).

In assessing the second aim of the study, the pattern of potentially traumatic events most commonly experienced, when not including “other events”, is consistent with the previous literature (Bohannon et al., 2019; Frazier et al., 2009; Phipps et al., 2019), with “receiving news about the death of a loved one” and “accidents” as being considered the most frequently reported events. Given the breadth and commonality of these event categories, it is unsurprising that these are reported as among the most frequently experienced across populations.

In relation to the study’s final aim, different categories of potentially traumatic events were associated with significant differences in PTS symptom severity. While event categories involving interpersonal abuse or an act of malice did not significantly differ from one another, there was a substantial difference in the severity of symptoms when compared to accidental events, or those caused by natural disasters. Specifically, higher levels of PTS symptoms were reported by individuals who had experienced intentional acts of aggression/abuse as compared to those who experienced non-intentional events.

When examining clinically significant PTSD symptomatology, 32% of those exposed to a potentially traumatic event indicated a level of symptom severity consistent with a likely diagnosis (22% of the total

sample). Rates of probable PTSD diagnosis were also higher among those who had experienced abuse/violence as children or adults (46%) as compared to those who experienced accidents, natural disasters, witnessed or received news of events (26%). The overall rate of clinically significant PTSD symptomatology is somewhat higher compared to population estimates of around 25% of individuals exposed to one or more traumatic event going on to experience PTSD (Hidalgo & Davidson, 2000). This may well be partly due to overestimation from self-report rather than clinician administered measures. Nevertheless, the present data suggest that a substantial proportion of the current sample experienced PTS symptoms at a level that causing potentially clinically significant distress. Conversely, our results also highlight the consistent finding in the broader literature, that while a significant proportion of students are exposed to potentially traumatic events (risk), not all develop severe distress symptoms. This distinction is crucial for developing targeted interventions that address both the prevention of distress in at-risk individuals and the treatment of actual distress in those affected.

The findings of the current study are consistent with those of Frazier et al. (2009) in suggesting that exposure to potentially traumatic events is relatively common among university undergraduates across Australian and US samples. The samples did, however, show some discrepancy in rates of events reported, with 64% of the current sample compared to 85% in the Frazier et al. (2009) sample reporting at least one event. In considering potential reasons for this discrepancy, it is possible that different screening instruments may have captured different event types. While Frazier et al. (2009) employed the 22-item Traumatic Life Events Questionnaire (Kubany, 2004) in the current study, we used the 11-item TEQ. While it is possible that the larger number of events covered by the Traumatic Life Events Questionnaire prompted the identification of more events in the Frazier et al. (2009) study, this scale focuses on more discrete events, while the TEQ captures broader event categories. Also, both scales ask about the possible experience of other potentially traumatic events meaning that they should both comprehensively capture potentially relevant experiences. It is possible, therefore, that the discrepancy across studies reflects genuine differences in the rate at which these events are experienced across these populations.

The higher rates of psychological distress observed among those exposed to potentially traumatic events underscore the additional challenges that may be faced by undergraduates with a history of trauma

experience. The associated emotional and cognitive disruptions that are common among those with a trauma history (Malarbi et al., 2017; Perfect et al., 2016; Van Der Kolk, 2007) suggests that periods of combined stress and focused cognitive effort (assignments/exams) may be particularly difficult for those with higher levels of PTS symptoms. To mitigate against such barriers, university administrators may consider additional support and accommodations to ensure equity for those who may struggle with coursework as a result of past trauma.

While trauma can cause considerable distress, there is growing recognition of the potential for post-traumatic growth following such experiences. Post-traumatic growth refers to the positive psychological changes and personal growth that can emerge following highly challenging life circumstances like trauma (Jayawickreme et al., 2021). Future studies examining both the distressing impacts and growth experiences of students exposed to potentially traumatic events would offer a more comprehensive understanding. Exploring the pathways to resilience and growth can inform strength-based interventions that not only alleviate distress but also facilitate post-traumatic growth. Moreover, incorporating qualitative methods could provide rich, in-depth insights into the experiences of trauma-exposed students. Interviews or focus groups, can capture the complexity of individual trauma narratives, highlighting unique aspects of personal growth that may be overlooked in quantitative studies. These methods would allow for exploration of how students interpret and make meaning from their traumatic experiences, thereby offering valuable context to the quantitative findings and potentially guiding tailored interventions.

While contributing to existing literature about rates and type of potentially traumatic event exposure among undergraduates, it is necessary to note limitations associated with the current study. The present results are constrained to the population under examination and, as noted via differences in the rates of trauma experienced in similar international populations (Frazier et al., 2009), the generalisability of such findings cannot be assumed. The higher proportion of female participants also limited the ability to systematically examine gender differences in traumatic event experiences. Thus, wider sampling in future research with a focus on greater gender balance may be beneficial.

The present study allowed us to identify the frequency of exposure to potentially traumatic events across individuals in the sample. However, the screening measure employed did not allow us to establish

the frequency with which individual event subtypes occur within individuals (i.e., individuals who experience multiple events), nor the relative intensity of individual trauma events (e.g., experience of associated injury with accidents). While we were able to determine the overall occurrence of different potentially traumatic events in the sample, the attenuated screening version of the TEQ employed in the current study did not permit assessment of how often individuals experienced multiple distinct traumas or repeated occurrences of the same trauma type. As such, future research could usefully seek to employ the full version of the TEQ, including follow-up questions regarding the frequency and severity of each individual event which would yield more comprehensive trauma history measures providing valuable insights into the cumulative effects of multiple or chronic trauma exposures in this population.

It is also important to note the limitations of the cross-sectional nature of the present study. This is particularly noteworthy when considering findings showing that psychological distress may interact with and mutually reinforce personality traits such as neuroticism. In a study examining the reciprocal relationship between neuroticism and life stress, Metts et al. (2021) found that neuroticism increases both interpersonal and non-interpersonal stress, and that stress can, in turn, elevate levels of neuroticism over time. It is therefore possible that underlying personality factors could have influenced the current findings. For instance, individuals with higher neuroticism may both experience and report greater psychological distress following trauma, and the trauma itself may further exacerbate neuroticism, creating a cyclical pattern that cross-sectional data cannot fully capture. As such, it will also be useful for future research to include longitudinal examination of the associations between trauma status, psychological distress, and aspects of personality, in addition to critical indicators of performance at university such as grades and dropout rates. Such findings will assist in clarifying the reciprocal relationships involved and where the greatest need lies in relation to trauma-exposed individuals for and where universities should concentrate on support services.

In summary, this study found that trauma exposure is common among Australian university students, with 64% reporting experiencing at least one potentially traumatising event. The most prevalent trauma types were unexpected death of a loved one, adult abusive experiences, and childhood abusive experiences. Students who reported trauma exposure had significantly higher psychological

distress than non-exposed students. Certain trauma types, especially those involving interpersonal violence and abuse, were associated with greater post-traumatic stress severity. These findings reinforce that trauma experience and related mental health impacts are highly prevalent issues affecting a majority of students. Universities have an important role to play in trauma-informed care, through screening, counselling services, victim support programs, educator training, and building resilience-oriented communities. With greater awareness and properly implemented trauma-informed approaches, institutions can better support student wellbeing and academic success.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The current data (individual demographics removed) are available on open science framework via the following link: https://osf.io/9ja82/?view_only=11751bb9be4d4155b15ea2ae167f21a1.

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